

Modern privacy regulations such as the EU General Data Protection Regulation (GDPR), India’s Digital Personal privacy compliance, large language models, retrieval-augmented generation, cookie consent auditing, web telemetry,

Introduction

With the rise of data privacy laws worldwide, including the EU’s General Data Protection Regulation (GDPR) and India’s Digital Personal Data Protection Act (DPDPA), privacy compliance is becoming a critical business requirement. Despite these legal frameworks, real-world compliance is highly fragmented. Many organisations use template-based

OptMotivation

While automated tools have been developed to detect trackers or parse privacy policies, they typically operate in silos. The main research question addressed by the wider project, and motivated by this review, is: *How can privacy compliance be automated?*

OptPaper Organisation

The remainder of this paper is structured as follows. Section 2 presents an integrated literature review and gap analysis.

Literature Review and Identified Gaps

To build a sound academic foundation, a systematic search was conducted across IEEE Xplore, the ACM Digital Library, and Google Scholar.

OptNLP and Machine Learning on Privacy Text

Analysing unstructured legal text is the first hurdle in automated compliance checking. Wilson et al. [0] established a baseline for privacy policy analysis. *Gap identified:* Text-only NLP methods assess what a policy *says*, but cannot verify what a system actually *does* [0, 0, 0].

OptBehaviour-to-Policy Inconsistency Detection

A second line of work checks whether disclosures are honest by analysing software behaviour. Zimmeck et al. [0] proposed a framework for detecting inconsistencies between privacy policies and actual system behaviour. *Gap identified:* Hybrid notice-versus-practice auditing exists only for mobile ecosystems via static analysis, leaving dynamic web environments unaddressed.

OptDynamic Web Cookie and Consent Auditing

Dynamic auditing of web tracking has been addressed through large-scale crawling. Trevisan et al. [0] performed a large-scale audit of web cookies. *Gap identified:* Dynamic web auditing measures behaviour and consent signals but never reconciles them with the site’s privacy policy.

OptLLM-Based Legal and Compliance Evaluation

The most recent direction applies large language models to legal reasoning. Guha et al. [0] released LegalBench, a comprehensive benchmark for legal reasoning. *Gap identified:* LLMs can reason over legal text and policies, but no framework feeds them live behavioural telemetry to

	Ref	Focus	Key Contribution
[!t] Representative Related Work on Automated Privacy-Compliance Auditing	[0]	Policy-text corpus	OPP-115 annotated corpus; statistical classification of privacy practices
	[0]	Deep-learning policy analysis	Polisis CNN classifier (88.4%) and QA assistant
	[0]	Policy QA	PrivacyQA dataset; BERT span extraction
	[0]	Notice vs. practice (mobile)	Hybrid text + static bytecode analysis
	[0]	Compliance at scale	MAPS large-scale automated analysis
	[0]	Privacy-label non-compliance	Lalaine measure vs. behaviour gap
	[0]	Cookie-law auditing	Detects pre-consent tracking cookies
	[0]	Cookie-notice compliance	General large-scale consent compliance analysis
	[0]	LLM policy evaluation	LLM assessment of privacy clauses vs. 10 lawyers
	[0]	Legal-reasoning benchmark	LegalBench: 16 legal-reasoning tasks

OptKey Findings

Reading across the four directions yields three consolidated findings, each grounded in the reviewed evidence: Written privacy policies are frequently incomplete and ambiguous and fail to disclose active tracking behaviour, as established by Wilson et al. [0]. Dynamic consent mechanisms on the web frequently set tracking cookies before obtaining consent, constituting measurable violations of GDPR and DPDPA. Large language models can interpret legal clauses and assess policy text against codified law with growing accuracy, but no framework integrates this with live behavioural data.

OptResearch Gap

These findings expose a clear gap. Existing systems analyse the written policy text in isolation (NLP/LLM approaches) or track behaviour without contextualising it against the policy (dynamic auditing). A holistic framework that reconciles these two perspectives is needed.